

# No. of Functional Features Included in the Solution

Functional Features Overview: This document lists the functional features included in the ComicCraft project solution. The features are derived from the project description and implementation workflow, covering user input, AI story generation, image generation, preview, PDF export, API integration, and deployment.

## Functional Features

### Feature Summary:

### Feature Category Breakdown:

### Implementation Notes:

- ComicCraft uses FastAPI for backend routing and request processing.
- Gemini Flash is used for structured comic panel outlines, while Gemini Pro is used for detailed narration and dialogue.
- Stable Diffusion is used for comic-style image generation.
- Jinja2 templates provide dynamic rendering of the homepage, comic preview, and export-success pages.
- FPDF is used to compile the generated comic into a multi-page PDF.
- The current project focuses on single comic generation and does not include user account management or a database.

Team ID: SWTID-2026-9766

|               |                                                         |
|---------------|---------------------------------------------------------|
| Date          | 29 September 2026                                       |
| Team ID       | SWTID-2026-9766                                         |
| Project Name  | ComicCraft - AI Comic Story Creator using Gemini Models |
| Maximum Marks | 5 Marks                                                 |

| S.No | Feature Name                                  | Feature Description                                                                                          | Module / Component                     | Status | Marks | Contribution                                    |
|------|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------|----------------------------------------|--------|-------|-------------------------------------------------|
| 1    | Comic Prompt & User Input                     | Collects story prompt, character name, setting, story tone, and preferred art style.                         | index.html / Frontend                  | Done   | 0.60  | Enables personalized comic creation.            |
| 2    | Structured Comic Outline Generation           | Generates a structured 5-panel comic outline with panel number, title, scene description, and image prompt.  | gemini_flash.py / Gemini Flash         | Done   | 0.60  | Creates a consistent 5-panel story structure.   |
| 3    | Story Narration & Dialogue                    | Expands the panel outline into full comic narration and character dialogues.                                 | gemini_pro.py / Gemini Pro             | Done   | 0.60  | Provides complete narration and dialogue.       |
| 4    | AI Comic Illustration Generation              | Creates comic-style illustrations from image prompts using Stable Diffusion.                                 | image_generator.py / Stable Diffusion  | Done   | 0.60  | Adds visual illustrations for each panel.       |
| 5    | Comic Panel Layout & Preview                  | Matches generated images with panel stories and displays the complete comic panel-by-panel.                  | layout_builder.py / comic_preview.html | Done   | 0.60  | Combines visual and textual content for review. |
| 6    | PDF Comic Export                              | Compiles panel images, titles, descriptions, captions, and narration into a downloadable multi-page PDF.     | exporters.py / FPDF                    | Done   | 0.60  | Provides a shareable/offline comic output.      |
| 7    | FastAPI Routes & JSON API                     | Provides browser and API routes including /generate, /generate-comic/json, /export-success, and /test-image. | routes.py / FastAPI                    | Done   | 0.60  | Supports browser and direct API workflows.      |
| 8    | Dynamic Jinja2 Frontend & Export Confirmation | Dynamically renders input, preview, and export-success pages using Jinja2 templates.                         | Jinja2 / HTML / CSS                    | Done   | 0.60  | Provides a smooth dynamic user experience.      |

| Metric                             | Count / Value |
|------------------------------------|---------------|
| Total Features Planned             | 8             |
| Total Features Implemented         | 8             |
| Core / Must-Have Features          | 6             |
| Additional / Nice-to-Have Features | 2             |

|                            |   |
|----------------------------|---|
| Features Tested & Verified | 8 |
|----------------------------|---|

| S.No | Category                  | Features in Category | Example Features                                       |
|------|---------------------------|----------------------|--------------------------------------------------------|
| 1    | User Interface (UI)       | 2                    | User input form; comic preview; export success page    |
| 2    | Backend / Logic           | 3                    | Outline, story, image and layout processing            |
| 3    | Database / Storage        | 0                    | No database/user-account management in current version |
| 4    | API / Integration         | 2                    | Gemini API; Stable Diffusion; FastAPI JSON endpoint    |
| 5    | Security / Authentication | 1                    | API keys handled through environment variables         |